

Adaboost regression

Problem: House price prediction

House	Area	Actual price
A	1000	100
B	1200	120
C	1500	150
D	1800	180
E	2000	250

Notice E is unusual $\rightarrow$ Most house follow

Area $\uparrow \rightarrow$ Price $\uparrow$

But E is much more expensive

Step 1: Initialize weights

$$w_i = \frac{1}{N}$$

$$w_i = 0.2$$

House	weights
A	0.2
B	0.2
C	0.2
D	0.2
E	0.2

Step 2: Regression stump

Area $< 1700 \rightarrow 123$

Area $\geq 1700 \rightarrow 215$

House	Actual	Predicted
A	100	123
B	120	123
C	150	123
D	180	215
E	250	215

Step 3: Calculate Absolute Error

$$e_i = |y_i - \hat{y}_i|$$

House	Error
A	23
B	3
C	27
D	35
E	35

Maximum Error = 35

Step 4: Normalize Error

$$L_i = \frac{e_i}{E_{\max}}$$

House	L_i
A	0.657
B	0.086
C	0.771
D	1.0
E	1.0

Step 5: Compute Average weighted loss

$$= 0.2 (0.657 + 0.086 + 0.771 + 1 + 1)$$

$$= 0.703$$

Step 6: Compute Beta

$$\beta = \frac{\bar{L}}{1 - \bar{L}} = \frac{0.703}{0.297} = 2.37$$

This is poor regressor because loss is high

Real adaboost require < 0.5 to continue

I am keeping numbers simple for learning

$$\bar{L} = 0.30$$

$$\beta = 0.429$$

Step 7: Update weights

A $\Rightarrow$

B $\Rightarrow$

E $=$

$$w_{\text{new}} = w_i B^{(1-L_i)}$$

$$0.2 \times 0.429^{(1-0.857)} = 0.15$$

$$0.2 \times 0.429^{(1-0.86)} = 0.09$$

$$0.2 \times 0.429^0 = 0$$

After Normalization

House

Weights

A

0.21

B

0.12

C

0.23

D

0.22

E

0.22

B $\Rightarrow$ predicted well $\rightarrow$ weight dropped

D & E were difficult $\rightarrow$ weight stay high

Now using bin we select weight which are high more times $\rightarrow$ as in regressor & repeat the process (with updated weights)

Note: In classification also updated weights from previous stump are used

Prediction

$$a_t = \ln\left(\frac{1}{\beta_t}\right)$$

$$a_1 = 0.846$$

$$a_2 = 0.663$$

$$a_3 = 1.266$$

Bay Area ≈ 1600

Stump

Prediction

1

155

2

155

3

160

Weighted median
 Sort prediction

Stump	Prediction	Weight	Cum Weight
2	155	0.663	0.663
3	160	1.266	1.929
1	165	0.846	2.775

Total weights = 2.775

Half weight = 1.3875

Crosses Half at 160

Final prediction = 160